In your working group, Department of Computer Science

Summative Coursework Front Page

Module Title	Software Engineering and Professional Development
Module Code	CS2SE
Lecturer responsible	Dr Lily Sun and Dr Nisha Singh
Coursework Title / short description	The SE Project Stage 1 assignment aims to apply Software Engineering principles, paradigms, and architectural approaches in the development of a Hospital Management System.
Type of Assignment (select one and delete rest)	Written Coursework Assignment
Learning Outcomes Assessed	See “Assessment Classification for the Learning Outcomes” on page 2
Individual or Group Assignment	Group assignment
Weighting of the Assignment	50%
Eligible for Self-Certification Extension (Yes/No)	No! However, all group members can independently apply for AA if needed.
Word count/Page limit or Test/Presentation Duration	7 pages (font: Calibri; font size: 11; line spacing: at 14), + one extra page for Effort Sheet
Typical expected hrs spent for the assignment	20 hours
Items to be submitted	A singly report in PDF (including the effort sheet) to be submitted to Blackboard
Work to be submitted on-line via Blackboard Ultra by	16 November 2026
Work will be marked and returned by	15 working days after the submission deadline
Artificial Intelligence Tools	May be used to support work

Note

By submitting this work, you are certifying that you have read the assessment guidelines, which are displayed in the folder of Assessment on the Blackboard course for this module, and that you have conformed to and understand the associated policies and practices, including those on:

- submitting your own work, rather than work produced by other people or systems (including those using artificial intelligence), and the associated penalties for Academic Misconduct;
- submitting your work by the specified deadline and the penalties associated with late submission, where applicable;
- complying with the requirements of the Assessment Adjustments (AA) system, where applicable; and
- attaching a green sticker for relevant needs to your submission to indicate that you have an approved adjustment.

1. Assessment classification for the Learning Outcomes

Classification	#1 Software Architecture & Solutions	#2 Problem-solving & Integration	#3 Evaluation & Implications
>80% (Excellent)	Demonstrates exceptional mastery of software architecture, modern development methodologies, and software engineering principles. Designs and implements highly reliable, scalable, adaptable, and innovative software solutions that effectively support complex decision-making scenarios. Integrates best practices and advanced patterns, anticipating future evolution and maintenance needs.	Expertly applies problem-solving frameworks, management tools, and strategic approaches. Seamlessly develops and integrates business processes, technological infrastructures, and emerging technologies. Solutions are fully aligned with organisational objectives, demonstrating strategic foresight and optimisation.	Critically evaluates system quality, safety, and trade-offs with exceptional depth. Considers all relevant legal, ethical, social, security, and economic implications in digital transformation initiatives. Provides well-justified, evidence-based recommendations to enhance outcomes.
70% - 79% (Very good)	Shows strong understanding and application of architecture, methodologies, and principles. Designs reliable, scalable, and adaptable solutions that support decision-making in most scenarios. Applies appropriate design patterns and practices, with minor omissions in foresight or maintainability.	Applies frameworks, tools, and strategies competently. Integrates business processes, technologies, and emerging trends effectively, with only minor alignment gaps. Demonstrates a good understanding of organisational objectives and strategic considerations.	Evaluates system quality, safety, and trade-offs thoroughly. Considers most relevant legal, ethical, social, security, and economic implications. Recommendations are well-informed and practically applicable.
60% - 69% (Good)	Demonstrates good understanding of architecture and methodologies. Implements solutions that are generally reliable and scalable, but may have minor shortcomings affecting adaptability or decision support. Applies common design patterns with moderate proficiency.	Uses frameworks, tools, and strategies adequately. Integration of processes and technologies is generally effective, though some gaps in alignment or optimisation exist.	Identifies system quality issues and trade-offs, considering some legal, ethical, social, security, and economic implications. Recommendations are generally sound but may lack depth or nuance.
50% - 59% (Satisfactory)	Demonstrates basic understanding of architecture and methodologies. Solutions are functional but may lack reliability, scalability, or adaptability. Limited use of design patterns or best practices.	Uses frameworks, tools, and strategies in a limited way. Integration of processes and technologies is partial, with gaps in alignment with organisational objectives.	Evaluates only basic aspects of system quality and trade-offs. Awareness of legal, ethical, social, security, and economic implications is limited. Recommendations are superficial or incomplete.
40% - 49% (Pass)	Shows minimal understanding of architecture or methodologies. Solutions are unreliable, non-scalable, or inappropriate, offering weak decision-making support. Rarely applies design principles effectively.	Very limited application of frameworks, tools, or strategies. Integration of processes and technologies is incomplete or ineffective. Strategies poorly support organisational objectives.	Provides minimal evaluation of quality, trade-offs, or implications. Consideration of legal, ethical, social, security, and economic factors is superficial.
<40% (Fail)	Fails to demonstrate understanding of architecture or methodologies. Solutions are non-functional, inappropriate, or unsafe. No application of design principles.	Fails to apply frameworks, tools, or strategies. Integration of processes and technologies is absent or ineffective. No alignment with organisational objectives.	Fails to evaluate system quality, safety, or trade-offs. Ignores legal, ethical, social, security, and economic implications.

2. Effort allocation sheet

You are required to work on the coursework in a group of 4 members. Each group will submit an *effect allocation sheet* together with the single report.

Name	Clearly specify your individual contributions in the relevant sections and/or parts of the submission.	Please confirm your expected level of effort for the assignment (e.g., 100% or less).	Please indicate whether you have included your Individual Reflection section in the overall submission (Y/N).	Signature

Note:

1) It is expected that all group members contribute equally (100%) to the coursework. If this is not the case, please estimate the percentage of effort you contributed to your parts.
2) The overall group mark will be adjusted to calculate each member's individual mark as follows: Individual Mark = Overall Mark × Claimed Individual Effort (%).
3) Additionally, each group member must include a 200-word individual reflective statement as part of the single submission. Failure to provide a complete or satisfactory reflection may result in a deduction of up to 10% from the member's original mark.

3. Assignment submission requirements

Your report is suggested to adopt the structure outlined as follows:

Module Code:	CS2SE
Assignment Report Title:	SE project stage 1 report
Student Number (e.g. 25098635):	
Actual hrs spent for the assignment:	
Which AI tools used (if applicable) and brief justification in Individual Reflection section:	

1. **Introduction** (10 marks, half a page)
   a) Provide a concise introduction outlining the purpose of the report (e.g. the background of the chosen use case scenario).
   b) Outline your intended software system context and discuss the crucial role of software paradigms and software architecture of microservices in supporting its development.

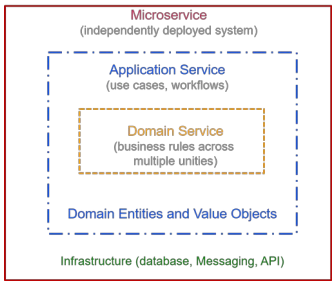
2. **Software Engineering: Principles, Paradigms, and Architecture**

2.1 Role of Software Paradigms in Designing the chosen use case scenario (20 marks, 2 pages)

   a) Examine how the selected software paradigm influenced the development of your intended software system, drawing on insights and enhanced outcomes from the weekly practical exercises.
   b) Justify the evaluation results from 2.1(a) to demonstrate that the chosen paradigm meets the intended expectations.

2.2 The development of a microservice for the chosen use case scenario (40 marks, up to 4 pages)

   a) Desing a microservice, following the provided design template, based on one of the recommended use case scenarios outlined in Appendix 1.
   b) Embed social, security, and ethical considerations into the microservice architecture.
   c) Justify design decisions by demonstrating the design of Domain Services, Application Services, Domain Entities, Value Objects, and Infrastructure components.



3. **Conclusion and recommendations** (10 marks, half a page)
   a) Draw concluding remarks (i.e., say what has said in the report); and
   b) Propose two significant recommendations for future improvements in adopting the software paradigm and refining your intended software system architecture.

- **Presentation of the report** (20 marks)

A suitable writing style for technical reporting with the group effort, clear and logical structure to support discussions/justifications, appropriate used of references, citations, and captions (see Appendix 2).

** **Individual Reflection on learning process and experience** (200 words per member – excluded from the page limit; referring to the point 3 of 'Note' on page 3 regarding marking)

- Comments on the group dynamics, planning, coordination, collaboration and role played in the team throughout the process of the analysis and design; and
- Your reflection on your contribution, experience gained, and lessons learned about formative assessment supported the learning, and GenAI aided skill development (if you engaged with those and used ethically).

Sub-total: 100 Marks

Appendix 1 Recommended use case scenarios

This list of use case scenarios supports the Task 2.2 in this CW.

1. online shopping platform
2. university student management system
3. online learning platform
4. hospital management systems
5. Telemedicine platform
6. digital banking platform
7. mobile payment application
8. insurance claim systems
9. Ride-share platform
10. food delivery platform
11. smart packing management systems
12. smart energy monitoring application
13. hotel booking systems
14. others

Appendix 2 Supporting information to the coursework

Report formatting requirements

The report is suggested to use the writing style and format as follows:

- Discussions in the technical report
  a) focusing on technical aspects
  b) using technical terms
  c) containing relevant content succinctly
  d) attach ALL members' individual reflection (200 words x No. of members in a group) in this single submission document

- Heading with label – e.g. 1. Introduction
- Page number
- Front: Calibri
- Front size: 11
- Line spacing: 1.5 lines

- Using in-text citation and caption for figures/tables
  In-text citation for references:
  … (Booch, et al., 1999). more than two authors
  or
  Booch, et al. (1999) state …

  … (Smith and Davids, 2019). one or two authors

  In-text citation and caption for figures/tables:

Table 1: The testing results …	place it on top of the table
Table in-text citation	e.g. Table 1 presents … or … (Table 1).
Figure 1: The class model for …	place it below the figure
Figure in-text citation	e.g. Figure 1 shows … or … (Figure 1).

Reference formatting

Booch G., Rumbaugh J, and Jacobson I. (1999) *The Unifies Modelling Language User Guide*, Addison-Wesley, ISBN: 0-201-57168-4

Dennis A., Wixom B.H. and Tegarden D. (2005) *Systems Analysis and Design with UML - an object-oriented approach*. (2nd edition), John Wiley & Son Inc., ISBN: 0-471-65920-7

Using references where needed